

Equations for shear 3PCF

RAFAEL CHARLES HERINGER GOMES¹

¹ *University of Pennsylvania*

1. OVERVIEW

Our code implements a calculation of the eight shear three-point functions given a set of cosmological parameters. The computation can be understood as divided into three steps, for which we will present the modeling equations.

- Computation of the matter bispectrum.
- Computation of the convergence bispectrum.
- Integration of the shear 3PCF.

2. MATTER BISPECTRUM

For the Bihalofit computation of the bispectrum, we calculate the following from CLASS:

- The linear matter power spectrum $P(k, z)$ on a grid of k and z values.
- The non-linear scale k_{nl} as for each redshift value.
- The comoving distance $\chi(z)$ and the derivative $dz/d\chi(z)$.

Next, we compute the fitting parameters, as functions of $\log \sigma_8$ and of n_{eff} . We divide the parameters into global and dependent parameters, the former being k -independent, and the latter k -dependent. The global parameters are:

$$\begin{aligned}
 \log b_n &= -3.428 - 2.681 \log \sigma_8 + 1.624 \log \sigma_8^2 - 0.095 \log \sigma_8^3 \\
 \log c_n &= 0.159 - 1.107 n_{\text{eff}} \\
 \log \gamma_n &= 0.182 + 0.57 n_{\text{eff}} \\
 \log f_n &= -10.533 - 16.838 n_{\text{eff}} - 9.3048 n_{\text{eff}}^2 - 1.8263 n_{\text{eff}}^3 \\
 \log g_n &= 2.787 + 2.405 n_{\text{eff}} + 0.4577 n_{\text{eff}}^2 \\
 \log h_n &= -1.118 - 0.394 n_{\text{eff}} \\
 \log m_n &= -2.605 - 2.434 \log \sigma_8 + 5.710 \log \sigma_8^2 \\
 \log n_n &= -4.468 - 3.080 \log \sigma_8 + 1.035 \log \sigma_8^2 \\
 \log \mu_n &= 15.312 + 22.977 n_{\text{eff}} + 10.9579 n_{\text{eff}}^2 + 1.6586 n_{\text{eff}}^3 \\
 \log \nu_n &= 1.347 + 1.246 n_{\text{eff}} + 0.4525 n_{\text{eff}}^2 \\
 \log p_n &= 0.071 - 0.433 n_{\text{eff}} \\
 \log d_n &= -0.483 + 0.892 \log \sigma_8 - 0.086 \Omega_m \\
 \log e_n &= -0.632 + 0.646 n_{\text{eff}}
 \end{aligned} \tag{1}$$

In order to compute the dependent parameters, we sort each set of (k_1, k_2, k_3) to find the set $(k_{\text{textmin}}, k_{\text{mid}}, k_{\text{max}})$. Then, we define $r_1 \equiv k_{\text{min}}/k_{\text{max}}$ and $r_2 = (k_{\text{mid}} + k_{\text{min}} - k_{\text{max}})/k_{\text{max}}$. Now we compute the dependent parameters:

$$\begin{aligned}
 \log a_n &= -2.167 - 2.944 \log \sigma_8 - 1.106 \log \sigma_8^2 - 2.865 \log \sigma_8^3 - 0.310 r_1 \gamma_n \\
 \log \alpha_n &= \min(-4.348 - 3.006 n_{\text{eff}} - 0.5745 n_{\text{eff}}^2 + 10^{-0.9+0.2 n_{\text{eff}}} r_2^2, n_s) \\
 \log \beta_n &= -1.731 - 2.845 n_{\text{eff}} - 1.4995 n_{\text{eff}}^2 - 0.2811 n_{\text{eff}}^3 + 0.007 r_2
 \end{aligned} \tag{2}$$

From these parameters, we compute the one-halo term:

$$B_{1h}(k_1, k_2, k_3, z) = \prod_{i=1,2,3} \frac{1}{a_n q_i^{\alpha_n} + b_n q_i^{\beta_n}} \frac{1}{1 + (c_n q_i)^{-1}} \quad (3)$$

where $q_i = k_i/k_{nl}$.

The 2-halo and 3-halo effects are put together in the bihalofit model B_{3h} term. To compute this part, we take the F_2 kernel defined as

$$F_2(k_1, k_2, k_3) = \frac{5}{7} + \frac{2}{7k_1^2 k_2^2} \left(\frac{-k_3^2 + k_1^2 + k_2^2}{2} \right)^2 - \frac{-k_3^2 + k_1^2 + k_2^2}{2} \left(\frac{1}{2k_1^2} + \frac{1}{2k_2^2} \right) \quad (4)$$

Next, we compute the enhanced power spectrum:

$$P_{\text{enhanced}}(k_i, z) = \frac{1 + f_n q_i^2}{1 + g_n q_i + h_n q_i^2} P_L(k_i, z) + \frac{1}{m_n q_i^{\mu_n} + n_n q_i^{\nu_n}} \frac{1}{1 + (p_n q_i)^{-3}} \quad (5)$$

Finally, we get

$$B_{3h}(k_1, k_2, k_3, z) = 2(F_2(k_1, k_2, k_3) + d_n q_3) \prod_{i=1,2,3} \frac{1}{1 + e_n q_i} P_{\text{enhanced}}(k_1, z) P_{\text{enhanced}}(k_2, z) + \text{two permutations} \quad (6)$$

3. CONVERGENCE BISPECTRUM

We now compute the projected convergence bispectrum $B_\kappa(\ell_1, \ell_2, \ell_3)$. From the Limber approximation, we take $k_i \approx \ell_i/\chi(z)$. We load the selection function $n(z)$ of the source galaxies and write the lensing kernel as:

$$g(\chi) = \int_\chi^{\chi_{\text{max}}} d\chi' n(z(\chi')) \frac{\chi' - \chi}{\chi'} \quad (7)$$

The convergence bispectrum is now found by

$$B_\kappa(\ell_1, \ell_2, \ell_3) = \frac{27}{8} \left(\frac{H_0}{c} \right)^6 \Omega_m^3 \int_0^{\chi_{\text{max}}} d\chi \frac{1}{\chi} (1 + z(\chi))^3 g(\chi) B(\ell_1/\chi, \ell_2/\chi, \ell_3/\chi, z(\chi)) \quad (8)$$

4. COSMIC SHEAR THREE POINT CORRELATION FUNCTIONS

The natural shear components Γ^0 , Γ^1 , Γ^2 and Γ_3 , with the projection relative to the orthocenter, are computed by

$$\begin{aligned} \Gamma^0(x_1, x_2, x_3) &= \frac{2\pi}{3} \int_0^\infty \frac{d\ell_1 \ell_1}{(2\pi)^2} \int_0^\infty \frac{d\ell_2 \ell_2}{(2\pi)^2} \int_0^{2\pi} d\phi B_\kappa(\ell_1, \ell_2, \sqrt{\ell_1^2 + \ell_2^2 + 2\ell_1 \ell_2 \cos \phi}) \\ &\times e^{21\beta} ((e^{i(\phi_1 - \phi_2 - 6\alpha_3)}) J_6(A_3) + (e^{i(\phi_3 - \phi_2 - 6\alpha_1)}) J_6(A_1) + (e^{i(\phi_3 - \phi_1 - 6\alpha_2)}) J_6(A_1)) \end{aligned} \quad (9)$$

$$\begin{aligned} \Gamma^1(x_1, x_2, x_3) &= \frac{2\pi}{3} \int_0^\infty \frac{d\ell_1 \ell_1}{(2\pi)^2} \int_0^\infty \frac{d\ell_2 \ell_2}{(2\pi)^2} \int_0^{2\pi} d\phi B_\kappa(\ell_1, \ell_2, \sqrt{\ell_1^2 + \ell_2^2 + 2\ell_1 \ell_2 \cos \phi}) \\ &\times (e^{i(\phi_1 - \phi_2 + 2\phi_3)}) e^{2i(\beta - \phi - \alpha_3)} J_2(A_3) + (e^{i(\phi_3 - \phi_2)}) e^{-2i(\beta + \alpha_1)} J_2(A_1) + (e^{i(\phi_3 - \phi_1 - 2\phi_2)}) e^{2i(\beta + \phi - \alpha_2)} J_2(A_1) \end{aligned} \quad (10)$$

$$\Gamma^2(x_1, x_2, x_3) = \Gamma^1(x_2, x_3, x_1) \quad (11)$$

$$\Gamma^3(x_1, x_2, x_3) = \Gamma^1(x_3, x_1, x_2) \quad (12)$$

where the values for ϕ_i , β , A_i and α_i are given by:

$$\begin{aligned}
\cos \phi_1 &= -\frac{x_1^2 - x_2^2 - x_3^2}{2x_2x_3} \\
\cos \phi_2 &= -\frac{x_2^2 - x_3^2 - x_1^2}{2x_3x_1} \\
\cos \phi_3 &= -\frac{x_3^2 - x_1^2 - x_2^2}{2x_1x_2} \\
\cos 2\beta &= \frac{(\ell_1^2 + \ell_2^2) \cos \phi + 2\ell_1\ell_2}{\ell_1^2 + \ell_2^2 + 2\ell_1\ell_2 \cos \phi} \\
\sin 2\beta &= \frac{(\ell_1^2 - \ell_2^2) \sin \phi}{\ell_1^2 + \ell_2^2 + 2\ell_1\ell_2 \cos \phi} \\
A_1 &= \sqrt{(\ell_2x_3)^2 + (\ell_3x_2)^2 + 2\ell_2x_2\ell_3x_3 \cos \phi + \phi_1} \\
A_2 &= \sqrt{(\ell_1x_3)^2 + (\ell_3x_1)^2 + 2\ell_1x_1\ell_3x_3 \cos \phi + \phi_2} \\
A_3 &= \sqrt{(\ell_1x_2)^2 + (\ell_2x_1)^2 + 2\ell_1x_1\ell_2x_2 \cos \phi + \phi_3}
\end{aligned} \tag{13}$$

$$\begin{aligned}
\cos(\alpha_1) &= \frac{(\ell_2x_3 + \ell_3x_2) \cos \frac{\phi+\phi_1}{2}}{A_1} \\
\sin(\alpha_1) &= \frac{(\ell_2x_3 - \ell_3x_2) \sin \frac{\phi+\phi_1}{2}}{A_1} \\
\cos(\alpha_2) &= \frac{(\ell_3x_1 + \ell_1x_3) \cos \frac{\phi+\phi_2}{2}}{A_2} \\
\sin(\alpha_2) &= \frac{(\ell_3x_1 - \ell_1x_3) \sin \frac{\phi+\phi_2}{2}}{A_2} \\
\cos(\alpha_3) &= \frac{(\ell_1x_2 + \ell_2x_1) \cos \frac{\phi+\phi_3}{2}}{A_3} \\
\sin(\alpha_3) &= \frac{(\ell_1x_2 - \ell_2x_1) \sin \frac{\phi+\phi_3}{2}}{A_3}
\end{aligned}$$

The integrals can be rewritten in terms of new variables ψ and R . We write:

$$\psi = \arctan \frac{\ell_2}{\ell_1} \tag{14}$$

$$R = \sqrt{\ell_1^2 + \ell_2^2} \tag{15}$$

and find

$$\begin{aligned}
\Gamma^0(x_1, x_2, x_3) &= \frac{1}{6(2\pi)^3} \int_0^{2\pi} d\phi \int_0^{\pi/2} d\psi \sin 2\psi \int_0^\infty dR e^{2i\beta} J_6(R) T^0(x_1, x_2, x_3, \phi, \psi, R) \\
\Gamma^1(x_1, x_2, x_3) &= \frac{1}{6(2\pi)^3} \int_0^{2\pi} d\phi \int_0^{\pi/2} d\psi \sin 2\psi \int_0^\infty dR J_2(R) T^1(x_1, x_2, x_3, \phi, \psi, R)
\end{aligned} \tag{16}$$

where

$$\begin{aligned}
T^0(x_1, x_2, x_3, \phi, \psi, R) &= \frac{e^{i(\phi_2 - \phi_3 - 6\alpha_1)}}{A_1'^4} B_\kappa \left(\frac{R}{A_1'} \cos \psi, \frac{R}{A_1'} \sin \psi, \frac{R}{A_1'} \sqrt{1 + 2 \cos \psi \sin \psi \cos \phi} \right) \\
&+ \frac{e^{i(\phi_3 - \phi_1 - 6\alpha_2)}}{A_2'^4} B_\kappa \left(\frac{R}{A_2'} \cos \psi, \frac{R}{A_2'} \sin \psi, \frac{R}{A_2'} \sqrt{1 + 2 \cos \psi \sin \psi \cos \phi} \right) \\
&+ \frac{e^{i(\phi_1 - \phi_2 - 6\alpha_3)}}{A_3'^4} B_\kappa \left(\frac{R}{A_3'} \cos \psi, \frac{R}{A_3'} \sin \psi, \frac{R}{A_3'} \sqrt{1 + 2 \cos \psi \sin \psi \cos \phi} \right)
\end{aligned} \tag{17}$$

and

$$\begin{aligned}
T^1(x_1, x_2, x_3, \phi, \psi, R) = & \frac{e^{i(\phi_3 - \phi_2 - 2\alpha_1 - 2\beta)}}{A_1'^4} B_\kappa \left(\frac{R}{A_1'} \cos \psi, \frac{R}{A_1'} \sin \psi, \frac{R}{A_1'} \sqrt{1 + 2 \cos \psi \sin \psi \cos \phi} \right) \\
& + \frac{e^{i(2\phi - 2\alpha_2 + 2\beta + \phi_3 - \phi_1 - 2\phi_2)}}{A_2'^4} B_\kappa \left(\frac{R}{A_2'} \cos \psi, \frac{R}{A_2'} \sin \psi, \frac{R}{A_2'} \sqrt{1 + 2 \cos \psi \sin \psi \cos \phi} \right) \\
& + \frac{e^{i(-2\phi - 2\alpha_3 + 2\beta + \phi_1 - \phi_2 + 2\phi_3)}}{A_3'^4} B_\kappa \left(\frac{R}{A_3'} \cos \psi, \frac{R}{A_3'} \sin \psi, \frac{R}{A_3'} \sqrt{1 + 2 \cos \psi \sin \psi \cos \phi} \right)
\end{aligned} \tag{18}$$

with $A_i' = A_i/R$.

5. COORDINATE TRANSFORMATION

Or computed result for the Γ^i functions now can be transformed from orthocenter coordinates to centroid. The length of the line that bisects x_3 and goes to the vertex opposite x_3 is:

$$h_3 = \frac{\sqrt{2x_1^2 + 2x_2^2 - x_3^2}}{2} \tag{19}$$

The angle between this line and side x_3 is given by ψ_3 . We have:

$$\cos 2\psi_3 = \frac{(x_2^2 - x_1^2)^2 - 4x_1^2 x_2^2 \sin^2 \phi_3}{4h_3^3 x_3^3} \tag{20}$$

and

$$\sin 2\psi_3 = \frac{(x_2^2 - x_1^2)x_1 x_2 \sin \phi_3}{h_3^3 x_3^3} \tag{21}$$

which gives us

$$\exp 2i\psi_3 = \frac{(x_2^2 - x_1^2)^2 - 4x_1^2 x_2^2 \sin^2 \phi_3}{4h_3^3 x_3^3} + i \frac{(x_2^2 - x_1^2)x_1 x_2 \sin \phi_3}{h_3^3 x_3^3} \tag{22}$$

The other ψ_i angles are defined accordingly. Our new Γ^i functions can now be written as

$$\Gamma_C^0 = \Gamma_O^0 \exp 2i(\psi_1 + \psi_2 + \psi_3) \tag{23}$$

$$\Gamma_C^1 = \Gamma_O^1 \exp 2i(-\psi_1 + \psi_2 + \psi_3) \tag{24}$$

$$\Gamma_C^2 = \Gamma_O^2 \exp 2i(\psi_1 - \psi_2 + \psi_3) \tag{25}$$

$$\Gamma_C^3 = \Gamma_O^3 \exp 2i(\psi_1 + \psi_2 - \psi_3) \tag{26}$$

REFERENCES